

XII CBSE - BOARD - MARCH - 2018

CODE (91) SET - 4

Date: 23.03.2018

Computer Science - Solutions

SECTION - A

[Only for Candidates, who adopted for C++]

1. (a)

- Ans. (i) else : Keywords
(ii) Long : Keywords
(iii) 4Queue : Identifiers
(iv) _count : Identifiers

(b)

Ans. fstream.h
string.h

(c)

Ans.

```
typedef int Count;
void main ( )
{
    Count C;
    cout<<"Enter the count:";
    cin>>C;
    for (int K = 1 ; K<=C; K++)
        cout<< C << "*" << K <<endl;
}
```

(d)

Ans. 35 & 4

38 # 3

38 # 9

(e)

Ans. 40 : 55 : 46 : 61

(f)

Ans. 13@10@11@10@

2. (a)

Ans. void Execute (char A, int B) ; // Function 1
void Execute (int A, char B) ; // Function 2
void Execute (int P=10) ; // Function 3

(b)

Ans. (i) Single Inheritance
(ii) S.display () ;
S.First : : display () ;

(c)

Ans. Class CONTAINER
{ float Radius, Height;
int Type ;
float Volume ;
void CalVolume(float Radius, float Height, int n)
{Volume = (3.14*Radius*Radius*Height)/n;
}
void CalVolume(float Radius, float Height)
{Volume = 3.14*Radius*Radius*Height;
}

```
public :
    void GetValues()
    {cout << "Enter Radius";
      cin >> Radius;
      cout << "Enter Height";
      cin >> Height;
      cout << "Enter Type (1 for cone and 2 for cylinder)";
      cin >> Type;
      switch(Type)
      {case 1: CalVolume(Radius, Height, 3);
        ShowAll( );
        break;
       case 2: CalVolume(Radius, Height);
        ShowAll( );
        break;
      }
    }
    void ShowAll()
    {cout << "Radius : " << Radius;
      cout << "Height : " << Height;
      cout << "Volume : " << Volume;
    }
};
```

(d)

- Ans. (i) Multiple Inheritance
 (ii) DD,MM,YYYY, Title[30], Name[20]
 (iii) Start (), View (), Initiate (), Display (), Enter (), Show ()
 (iv) Schedule (), Course (), Teacher ()

3. (a)

Ans. void sumEO(int VALUES [], int N)
 {
 int S1 = 0;
 int S2 = 0;
 for(int i = 0 ; i < N ; i ++)

```

    { if(VALUEs[i]% 2 == 0)
      {S1 += VALUEs [i];}
    else
      {S2 += VALUEs [i];}
    }
    cout << "Sum of even values"<< S1 << endl;
    cout << "Sum of odd values"<< S2 << endl;
}

```

(b)

Ans. void UpperHalf(int Mat [4][4])
 {for (int i=0; i<4; i++)
 { for (int j=0; j<4; j++)
 { cout << Mat [i] [j] << " ";
 }
 cout << endl;
 }
 }

(c)

Ans. W - 2
 N - 20
 M - 15
 Loc(Data[15][10]) = 15000
 Row major formula :

$$\text{Loc}(\text{Data}[i][j]) = \text{Base}(\text{Data}) + W * (M * i + j)$$

$$\text{Loc}(\text{Data}[15][10]) = \text{Base}(\text{Data}) + 2 * (15 * 15 + 10)$$

$$15000 = \text{Base}(\text{Data}) + 2 * (225 + 10)$$

$$= \text{Base}(\text{Data}) + 2 * 235$$

$$= \text{Base}(\text{Data}) + 470$$

$$\text{Base}(\text{Data}) = 15000 - 470$$

$$= 14530$$

$$\text{Loc}(\text{Data}[10][5]) = 14530 + 2 * (15 * 10 + 5)$$

$$= 14530 + 2 * (150 + 5)$$

$$= 14530 + 2 * 155$$

$$= 14530 + 310$$

$$= 14840$$

(d)

```

Ans. void QUEUE :: AddPacket()
{
    Packet *temp = new Packet;
    if(temp==NULL)
    {
        cout<<"Overflow"<<endl;
    }
    cout<<"Enter PID";
    cin >> temp->PID;
    int n = temp->PID;
    temp -> Link=NULL;
    // for first node
    if(Front == NULL)
    {
        Front = Rear = temp;
    }
    else
    {
        Rear->Link = temp;
        Rear = temp;
    }
    cout<<n <<"has been inserted successfully."<<endl;
    //display
    if(Front==NULL)
    {
        cout<<"Underflow."<<endl;
        return;
    }
    else
    { temp = Front;
        //will check until NULL is not found
        while(temp)
        {
            cout << temp -> PID << " ";
            temp = temp -> Link;
        }
        cout<<endl;
    }
}

```

```

}
void QUEUE :: DeletePacket ()
{
    if(Front == NULL)
    {
        cout<<"underflow"<< endl;
        return;
    }
    cout << Front -> PID <<" is being deleted "endl;
    if(Front == Rear) // if only one node is there
        Front = Rear = NULL;
    else
        Front = Front -> Link;
    //display
    if(Front==NULL)
    {
        cout<<"Underflow."<<endl;
        return;
    }
    else
    {
        Packet *temp=Front;
        // will check until NULL is not found
        while(temp)
        {
            cout << temp -> PID <<" ";
            temp = temp -> Link;
        }
        cout<<endl;
    }
}
}

```

(e)

Ans.

Symbols	Stack	Expression
U		U
*	*	U
V	*	UV
+	+	UV*
C	+(	UV*
W	+(	UV*W
-	+(-	UV*W
Z	+(-	UV*WZ
)	+	UV*WZ-
/	+/	UV*WZ-
X	+/	UV*WZ-X
	+	UV*WZ-X/
		UV*WZ-X/+

Postfix expression : UV*WZ-X/+

4. (a)

Ans. void HashDisplay()

```
{
ifstream fin("MATTER.TXT");//opening text file
int size;
char ch;
fin.seekg(0, ios::end);
size = fin.tellg();
fin.seekg(0, ios::beg);
for(int i=0; i< size; i++)
{
fin.get(ch);
cout << ch << "#";
}
fin.close();
}
```

(b)

```
Ans. void putnot ( )
{
    cout << NOT << " ";
}

void Total Teachers( )
{
    fstream f;
    SCHOOLS Stu;
    f.open("SCHOOLS.DAT", ios::in | ios::binary);
    cout << "\nTotal no. of Teachers : \n";
    while( ( f.read( (char*) &Stu, sizeof(Stu))) != NULL )
        Stu.putnot();
    f.close();
}
```

(c)

Ans. Record : 7

SECTION C [For all the candidates]

5. (a)

Ans. (iv) CARTESIAN PRODUCT : VIDEO $\times$ MEMBER
Degree : 5
Cardinality : 9

(b)

Ans. (i) SELECT * FROM TRANSACT WHERE TYPE = 'Deposit'
(ii) SELECT ANO, AMOUNT FROM TRANSACT WHERE MONTH(DOT) = 10;
(iii) SELECT DOT FROM TRANSACT WHERE ANO = 103
ORDER BY DOT DESC LIMIT = 1
(iv) Select A.ANAME, T.ANO, T.DOT FROM ACCOUNT as A
JOIN TRASACT as T ON A.ANO = T.ANO
WHERE T.AMOUNT <= 3000;

(v)

ANO	ANAME
103	AliEReza
105	Simran Kaur

(vi)

ANO
101
103
102

(vii)

ANO	COUNT	MIN(AMOUNT)
101	2	2500
103	2	1000

(viii)

Count	Sum(AMOUNT)
2	5000

6. (a)

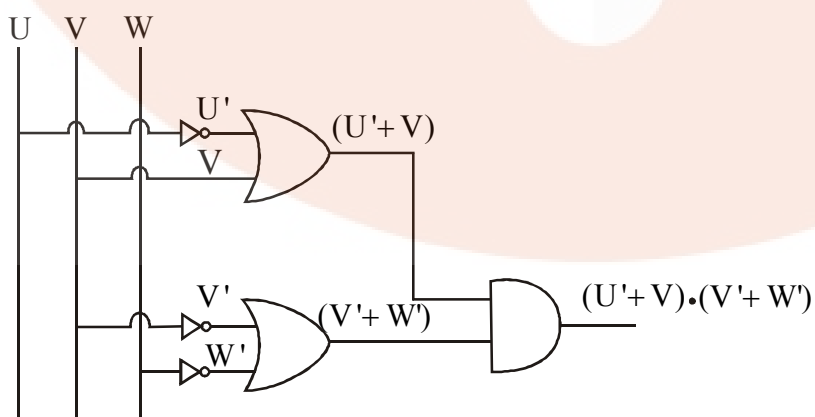
Ans. $X(X+Y) = X$

The truth is :

X	Y	XY	$X + X.Y$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

(b)

Ans.



(c)

Ans. Output : $(\bar{X} + \bar{Y} + Z)(X + \bar{Y} + \bar{Z})(\bar{X} + Y + \bar{Z})(\bar{X} + \bar{Y} + Z)$

(d)

Ans. $G(U, V, W, Z) = \Sigma(3, 5, 6, 7, 11, 12, 13, 15)$

$$= W.Z + V.Z + \bar{U}.V.W + U.V.\bar{W}$$

WZ \ UV	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	1	1	0
10	0	0	1	0

$\bar{U}.V.W$ (points to the cell at UV=01, WZ=11)
 $U.V.\bar{W}$ (points to the cell at UV=11, WZ=00)
 $V.Z$ (points to the cell at UV=01, WZ=11)
 $W.Z$ (points to the cell at UV=11, WZ=11)

7. (a)

Ans. A bus topology is a networking architecture that is linear, usually by using one or more pieces of cable to

form a single line, or bus. The Signals sent by one station extend the length of this cable to be heard by other stations.

A star topology is an architecture that includes a central device or hub to connect all stations together. Signals sent by a station must pass through (and are usually regenerated) by these central hubs.

Since the hub sits in the center and all other stations are linked through the hub, the architecture resembles a star.

Advantages of Star Topology

- As compared to Bus topology it gives far much better performance, signals don't necessarily get transmitted to all the workstations. A sent signal reaches the intended destination after passing through no more than 3-4 devices and 2-3 links.
- Centralized management. It helps in monitoring the network.

Disdvantages of Star Topology

- Too much dependency on central device has its own drawbacks. If it fails whole network goes down.
- The use of hub, a router or a switch as central device increases the overall cost of the network.

(b)

Ans. (i) Java Script : Client Side Scripting

(ii) ASP : Server Side Scripting

(iii) VB : Client Side Scripting

(iv) JSP : Server Side Scripting

(c)

Ans. (i) SMTP : Simple Mail Transfer Protocol

(ii) VoIP : Voice Over Internet Protocol

(iii) GSM : Global System for Mobile Communication

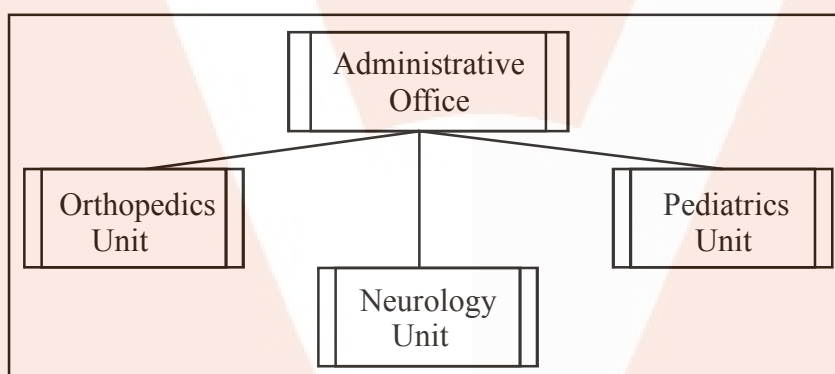
(iv) WLL : Wireless Local Loop

(d)

Ans. (i) **Administrative office** as this department contains the maximum number of computers thus decreasing

the cabling cost for most of the computers as well as increasing the efficiency of the maximum computers in the network.

(ii)



(iii) • **Switch** : Switch would be needed in each of these buildings to interconnect the computers installed within the building.

(iv) Topologies : Star Topology

Network Cable : Coaxial Cable